INCH-POUND
MIL-DTL-24643/30E
22 August 2002
SUPERSEDING
MIL-C-24643/30D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SWU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	Copper conductor, uncoated, ASTM B 286, AWG size No. 22,7 strands.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.048 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth -	Braided shield of AWG no. 36 uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Fifth -	Shield insulation of two polyester tapes, per NEMA FI 4, sealed. An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.
Sixth -	The required number of singles (see table I) cabled together with a lay not greater than 24 times the pitch diameter. Cabling sequence shall be consecutive, starting with No. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.
Seventh -	Binder tape, applied helically with overlap.

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Construction variant for type LS1SWU

Eighth - Cross-linked polyolefin jacket (see table I for thickness).

TABLE I. Details.

Military part no. M24643/30	Size	Number of conductors	Cable jacket thickness min avg (inch)	Overall diameter (inches) TYPE LS1SWU	
				min	max
-01UN	2	2	0.038	0.430	0.455
-02UN	14	14	0.050	0.825	0.870
-03UN	20	20	0.050	0.970	1.030
-04UN	30	30	0.063	1.135	1.200

EXAMINATION AND TESTS:

Basic electrical: Requirements:

Conductor resistance - ohms/1000 feet at 25°C, maximum 17.2
Voltage withstand - volts, root mean square, minimum
Conductor to shield 3000
Shield to shield 500
Insulation resistance - megohms -1000 feet minimum
Conductor to shield 500
Shield to shield 100
Conductor and shield continuity ..... No failure

Group A:

Visual and dimensional ..... No failure

Hydrostatic (open end) - leakage at 300 lb/in^2,
2 hours, in^3, maximum 1.0
Capacitance - 1 kilohertz (kHz), pF/ft, maximum 28
Characteristic impedance - at 1MHz, ohms ..... 61 $\pm$ 3

Group B:

Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (When tested at 200°C) 50
Drip - 95 $\pm$ 1°C ..... Zero
Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250

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Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

Group C:
Physicals (aged)
Insulation
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (jacket) - cycles, minimum 125
Cable sealant removability ..... No failure
Shield - conformance to material, construction and coverage ..... No failure

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Requirements:
Aging and compatibility (cable) (125 ± 5°C) ..... No failure
Abrasion resistance (jacket) - scrapes, minimum 75
Acid gas equivalent - percent, maximum
Jacket 2
Fillers 2
Insulation 18
Shield insulation 2
Halogen content - percent, maximum
Jacket 0.2
Fillers 0.2
Immersion (jacket)
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Smoke index, maximum

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Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-028)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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